

Improving Speech Recognition for Dysarthric Speech via Single-Utterance Test-Time Adaptation

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Abstract

Automatic Speech Recognition (ASR) systems achieve strong performance on standard speech benchmarks but often fail when applied to pathological speech such as dysarthria, stuttering, etc. This performance gap limits accessibility for individuals with speech impairments who rely on voice-driven technologies. In this work, we investigate lightweight adaptation strategies to improve ASR robustness for pathological speech without requiring large labeled datasets. We evaluate Single Utterance Test-Time Adaptation (SUTA), which adapts model parameters during inference using unsupervised objectives. Experiments on the TORGO dataset which contains speech from people with dysarthria show that SUTA improves word error rate (WER) for dysarthric speakers compared to zero-shot state-of-the-art models while maintaining competitive performance for control speakers. These findings highlight the potential of using inference-time adaptation to improve accessibility of ASR systems for underrepresented speech populations.

1 Introduction

Recent advances in Automatic Speech Recognition (ASR) have significantly improved transcription accuracy for standard speech. Large-scale models such as *Whisper* and self-supervised architectures such as *wav2vec 2.0* achieve strong performance on widely used benchmarks trained on thousands of hours of speech data. However, these models often fail when applied to speech produced by individuals with speech impairments.

Speech disorders such as dysarthria, stuttering, and speech impairments associated with neurological conditions including Parkinson’s disease introduce substantial variability in articulation, phonation, and speaking rate. As a result, speech signals produced by individuals with such conditions often deviate significantly from the acoustic patterns observed in typical speech datasets. This mismatch

leads to substantially higher word error rates when conventional ASR systems are applied to pathological speech.

A key challenge in developing robust ASR systems for speakers with disabilities is the limited availability of publicly accessible datasets. Clinical speech data is often highly sensitive and subject to strict privacy constraints, making it difficult to release at scale. While initiatives such as Project Euphonia have collected large amounts of speech from individuals with disabilities, many of these datasets remain closed or restricted-access. Consequently, open research on speech recognition for pathological speech remains underrepresented compared to standard ASR research.

The scarcity of data also introduces additional challenges when attempting to adapt ASR systems to speech impairments. First, the amount of available labeled speech is typically too small to support large-scale model training or fine-tuning. Second, speech produced by different individuals with the same disorder can vary substantially due to differences in severity, articulation patterns, and compensatory strategies. As a result, models adapted to one speaker often fail to generalize well to new speakers even when the underlying condition is similar.

One potential alternative to conventional fine-tuning is to leverage the zero-shot capabilities of large pre-trained ASR models. While larger models trained on massive speech corpora may demonstrate some robustness to unseen conditions, they often contain little or no speech from individuals with disabilities in their training data. Consequently, their performance on pathological speech remains limited, particularly for smaller and medium-sized models.

In this work, we explore whether lightweight adaptation techniques can improve ASR performance for pathological speech without requiring additional labeled data. Specifically, we investigate

Single Utterance Test-Time Adaptation (SUTA), a method that adapts a speech recognition model during inference using only the incoming utterance. SUTA updates a small subset of model parameters using self-supervised objectives. We specifically focus on the feature extractor and layer normalization parameters. Because this adaptation occurs at test time and does not require transcription labels, the method can adapt to new speakers using only a single utterance and without any additional training data.

We evaluate this approach on the TORGO dataset, a dysarthric speech corpus and compare SUTA-based adaptation against a state-of-the-art zero-shot model Whisper. Our results show that test-time adaptation significantly improves recognition performance for dysarthric speakers, particularly against smaller and medium-sized Whisper models.

2 Related Work

3 Methodology

3.1 Single Utterance Test-Time Adaptation

Speech from individuals with dysarthria often exhibits strong speaker-specific variability due to impaired articulation and motor control. Models trained on typical speech therefore face a distribution mismatch when applied to such speakers. To address this issue, we adopt *Single Utterance Test-Time Adaptation* (SUTA) (), a lightweight method that adapts model parameters during inference using only the incoming utterance.

Unlike traditional speaker adaptation methods that require labeled data or multiple utterances, SUTA performs unsupervised optimization on a single test utterance. During inference, the model parameters are updated to reduce uncertainty in the predicted output distribution, allowing the acoustic representations to better align with the characteristics of the current speaker.

This approach is particularly suitable for Connectionist Temporal Classification (CTC) based ASR models, which produce frame-level probability distributions over output tokens. These distributions allow unsupervised objectives such as entropy minimization to be computed without requiring ground-truth alignments.

3.2 Adaptation Objective

To perform unsupervised adaptation, we minimize the entropy of the model’s output distribution. Intu-

itively, a well-adapted model should produce confident predictions for the observed speech signal. Entropy minimization therefore encourages the model to sharpen its predictions for the current utterance.

The entropy loss is defined as:

$$L_{EM} = -\frac{1}{L} \sum_{i=1}^L \sum_{j=1}^C P_{ij} \log P_{ij} \quad (1)$$

where L is the number of acoustic frames and P_{ij} denotes the predicted probability of class j at frame i .

To prevent degenerate solutions in which the model collapses to a small set of output tokens, we additionally incorporate a Minimum Class Confusion (MCC) regularization term that discourages correlations between predicted classes.

$$L_{MCC} = \sum_{j=1}^C \sum_{j' \neq j} P_{.j}^T P_{.j'} \quad (2)$$

where $P_{.j}$ represents the vector of predicted probabilities for class j across all frames. This objective encourages the model to produce more discriminative predictions across classes.

The final adaptation objective combines these terms:

$$L = \alpha L_{EM} + (1 - \alpha) L_{MCC} \quad (3)$$

3.3 Parameter Selection for Adaptation

Rather than updating all model parameters, we restrict adaptation to a small subset of layers. Specifically, we update the layer normalization parameters and the feature extractor.

Layer normalization parameters are particularly suitable for test-time adaptation because they control the scaling and shifting of intermediate feature representations. Adjusting these parameters allows the model to recalibrate internal activations for speaker-specific acoustic characteristics without significantly altering the learned linguistic representations.

We also explore adapting the feature extractor, which operates directly on the acoustic signal. Since dysarthric speech often involves atypical articulation patterns and spectral distortions, allowing the feature extractor to adapt can help the model better capture speaker-specific acoustic cues.

By restricting adaptation to these components, we achieve lightweight personalization while maintaining the stability of the pretrained model.

4 Experimental Setup

4.1 Datasets

The TORGO dataset is a speech corpus that includes both acoustic and articulatory data from speakers with dysarthria, a motor speech disorder that affects the articulation of speech. It contains recordings from a variety of dysarthric speakers, making it a valuable resource for training models that can handle non-standard speech patterns.

The TORGO corpus includes a total of 19 speakers, divided into two categories: control speakers (healthy speakers) and non-standard speakers (those with dysarthria). Specifically, the dataset includes 3 female non-standard speakers, 3 female control speakers, 5 male non-standard speakers, and 4 male control speakers. This variation in both gender and speech condition provides a diverse set of recordings, enabling researchers to study how ASR models can be adapted and generalized to handle speech from individuals with varying levels of speech impairments.

In addition to the speaker categories, the dataset also provides information about the severity of the dysarthria for each non-standard speaker. The severity levels are as follows:

- Very Low Severity: F03, F04, M03
- Low Severity: F01, M05
- Medium Severity: M01, M02, M04

4.2 Evaluation Metrics

The primary evaluation metric for this study is Word Error Rate (WER). WER is a commonly used metric in speech recognition tasks, which evaluates the accuracy of transcriptions produced by an ASR model by comparing the predicted text with the reference transcription. The formula for WER is given by:

$$WER = \frac{S + D + I}{N}$$

where:

- S is the number of substitutions,
- D is the number of deletions,
- I is the number of insertions,
- N is the total number of words in the reference transcription.

Model	F01	F03	M01	M02
Whisper-tiny	108.0	61.0	316.0	341.0
Whisper-medium	86.0	57.0	69.0	236.0
Whisper-large-v3	80.0	45.2	76.0	81.0
SUTA (TTA-10)	87.05	49.04	110.12	86.20

Table 1: WER (%) comparison on non-standard speakers across Whisper models and SUTA.

4.3 Implementation Details

In this experiment, the model used for Automatic Speech Recognition (ASR) is Wav2Vec2ForCTC with a sample rate of 16,000 Hz. The model is pre-trained using the facebook/wav2vec2-base-960h checkpoint.

The optimizer used is AdamW, and the learning rate is set to $2e-5$ for this experiment. The model employs the temperature scaling parameter, set to 2.5, to adjust the output logits, improving the model’s performance in handling ambiguous cases during inference.

5 Experiments and Results

We evaluate the effectiveness of Single Utterance Test-Time Adaptation (SUTA) on dysarthric speech from the TORGO dataset. Performance is measured using Word Error Rate (WER), where lower values indicate better transcription accuracy. We compare SUTA with zeroshot Whisper models of varying sizes.

5.1 Comparison with Zeroshot Whisper Models

Tables 1 and 2 compare the performance of SUTA with Whisper-tiny, Whisper-medium, and Whisper-large-v3 on non-standard (pathological) and control speakers.

5.1.1 Non-standard Speakers

For speakers with dysarthric speech, SUTA significantly improves over the smallest Whisper model and performs competitively with Whisper-medium in several cases. The improvements are particularly noticeable for speakers with moderate impairment (e.g., F03).

While Whisper-large-v3 achieves the best overall performance due to its larger capacity and extensive pretraining, SUTA substantially narrows the performance gap using a much smaller adaptation mechanism. This suggests that test-time adaptation can partially compensate for the mismatch between pretrained models and pathological speech.

Model	FC01	FC02	MC01	MC02
Whisper-tiny	44.0	31.0	37.0	40.0
Whisper-medium	29.0	16.0	22.0	25.0
Whisper-large-v3	26.0	14.0	18.0	23.0
SUTA (TTA-10)	32.94	24.56	30.23	31.40

Table 2: WER (%) comparison on control speakers across Whisper models and SUTA.

TTA Steps	F01	F03	F04	M01	M02
Original	89.57	51.8	25.5	108.1	91.1
TTA-1	89.93	49.26	24.6	107.29	88.02
TTA-3	87.41	49.04	23.2	108.5	86.57
TTA-5	88.13	48.62	23.0	108.91	85.48
TTA-10	87.05	49.04	23.0	110.12	86.2

Table 3: WER (%) across different TTA steps for non-standard speakers.

TTA Steps	FC01	FC02	FC03	MC01
Original	38.58	28.68	34.11	36.97
TTA-1	34.12	24.88	31.65	33.09
TTA-3	32.64	25.2	30.49	31.2
TTA-5	32.34	24.88	30.1	32.14
TTA-10	32.94	24.56	30.23	31.4

Table 4: WER (%) across different TTA steps for control speakers.

TTA Step	F03		F04	
	With FE	No FE	With FE	No FE
Original	51.80	51.80	25.53	25.53
TTA-1	49.26	51.59	24.59	25.76
TTA-3	49.04	51.17	23.19	25.29
TTA-5	48.62	50.74	22.95	25.06
TTA-10	49.04	50.32	22.95	24.82

5.1.2 Control Speakers

For control speakers, Whisper models already achieve relatively low WER due to their training on typical speech. In this setting, SUTA improves over Whisper-tiny but does not outperform the larger Whisper-medium and Whisper-large models. This indicates that the benefits of test-time adaptation are more pronounced when the speech signal deviates from the training distribution.

5.2 Effect of Test-Time Adaptation Steps

To analyze the impact of adaptation steps, we evaluate the model at different TTA iterations.

5.2.1 Non-standard Speakers

As shown in Table 3, the largest improvements occur within the first few adaptation steps. After approximately five steps, the gains begin to stabilize, suggesting diminishing returns from additional adaptation. This indicates that only a small number of adaptation iterations are sufficient to personalize the model for a given speaker.

5.2.2 Control Speakers

For control speakers (Table 4), TTA also yields modest improvements, though the magnitude is smaller compared to non-standard speakers. This suggests that adaptation primarily benefits speech that differs significantly from the model’s training distribution.

5.3 Effect of Feature Extractor Adaptation

Finally, we examine the effect of updating the feature extractor during SUTA. Tables 5 compare results obtained with and without feature extractor (FE) adaptation.

Updating the feature extractor generally improves performance, particularly for speakers with stronger acoustic deviations (e.g., M02). This suggests that adapting early acoustic representations can help the model better capture speaker-specific characteristics in dysarthric speech.

6 Discussion

6.1 Adaptation to Atypical Speech Variability

Speech produced by individuals with dysarthria often exhibits substantial variability in articulation, timing, and acoustic realization. As a result, models trained primarily on typical speech distributions frequently struggle to generalize to such speakers. Our results demonstrate that single utterance test-time adaptation can partially mitigate this mismatch by allowing the model to adjust its parameters to the acoustic characteristics of the incoming speech signal.

The improvements observed for several non-standard speakers suggest that even a small amount of adaptation can help the model recalibrate its internal representations to better handle atypical pronunciation patterns. This is particularly important in clinical speech settings, where large-scale labeled data for individual speakers is rarely available. In such scenarios, lightweight adaptation techniques such as SUTA provide a practical mechanism for improving recognition performance without requiring additional supervision.

6.2 Rapid Personalization Through Test-Time Adaptation

An important finding from our experiments is that most of the performance gains occur within the first few adaptation steps. After approximately three to

TTA Step	M01		M02	
	With FE	No FE	With FE	No FE
Original	108.10	108.10	91.10	91.10
TTA-1	107.29	107.69	88.02	90.93
TTA-3	108.50	107.29	86.57	91.11
TTA-5	108.91	107.69	85.48	90.56
TTA-10	110.12	106.88	86.20	91.11

Table 5: Effect of feature extractor adaptation during SUTA.

five iterations, improvements tend to stabilize, indicating diminishing returns from additional updates.

This behavior suggests that the model is able to rapidly adjust to speaker-specific characteristics using only the information contained within a single utterance. From a deployment perspective, this is particularly encouraging: it implies that meaningful personalization can be achieved with minimal computational overhead and without requiring prolonged adaptation phases.

For assistive speech technologies used by individuals with speech impairments, such rapid adaptation could enable systems to continuously refine their behavior as new utterances are observed, gradually improving usability over time.

6.3 Role of Acoustic Representation Adaptation

Our ablation experiments further reveal that adapting the feature extractor can provide additional gains, particularly for speakers with higher severity. Feature extractors operate directly on the acoustic signal and therefore encode low-level information such as spectral structure and temporal dynamics. Allowing these layers to adapt enables the model to adjust its earliest representations to better reflect the acoustic properties of dysarthric speech.

6.4 Implications for Clinical Speech Technologies

Taken together, our results suggest that lightweight test-time adaptation offers a promising pathway for improving ASR performance in clinical speech applications. Unlike traditional speaker adaptation approaches, which require curated labeled datasets or extensive fine-tuning, SUTA can operate using only the incoming speech signal itself.

This property is particularly valuable in assistive communication systems, where collecting large quantities of labeled speech from individuals with speech disorders is often impractical. By enabling on-the-fly personalization, test-time adaptation could help bridge the performance gap be-

tween typical and atypical speech, ultimately improving accessibility and usability for speakers with speech impairments.

More broadly, these findings highlight the importance of adaptive modeling approaches for speech technologies deployed in real-world clinical contexts, where variability across speakers is the norm rather than the exception.

7 Conclusion

We investigated lightweight adaptation techniques to improve automatic speech recognition for dysarthric speech. Our experiments show that Single Utterance Test-Time Adaptation significantly improves recognition accuracy without requiring labeled adaptation data. These findings highlight the potential of inference-time adaptation to build more inclusive ASR systems.

8 Limitations

This work has several limitations. First, the TORGO dataset is relatively small, limiting the diversity of speech conditions studied. Second, our experiments focus on English speech only. Future work should evaluate these methods on larger and more diverse pathological speech datasets.

References